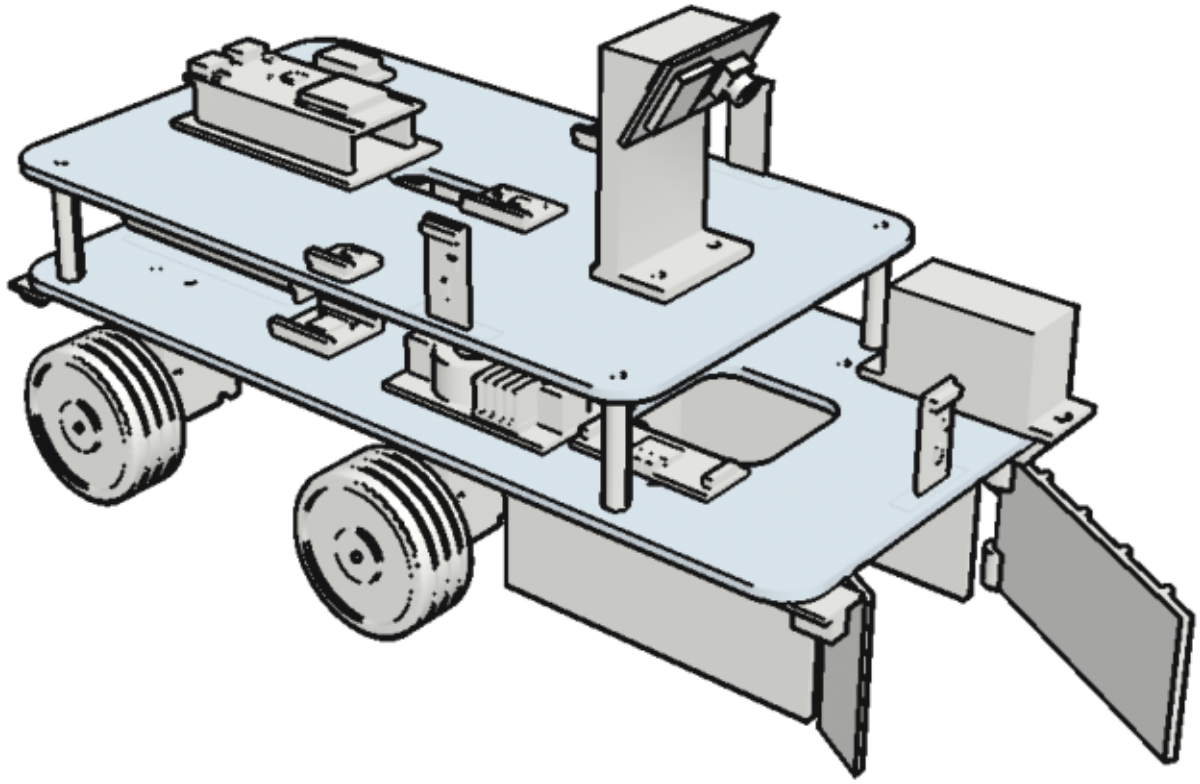
A.E.G.I.S. Mk2

Assembly guide



BEFORE YOU START

Tools: Phillips screwdriver, side cutters, wire strippers, soldering iron, multimeter, 2.5 mm hex key (optional).

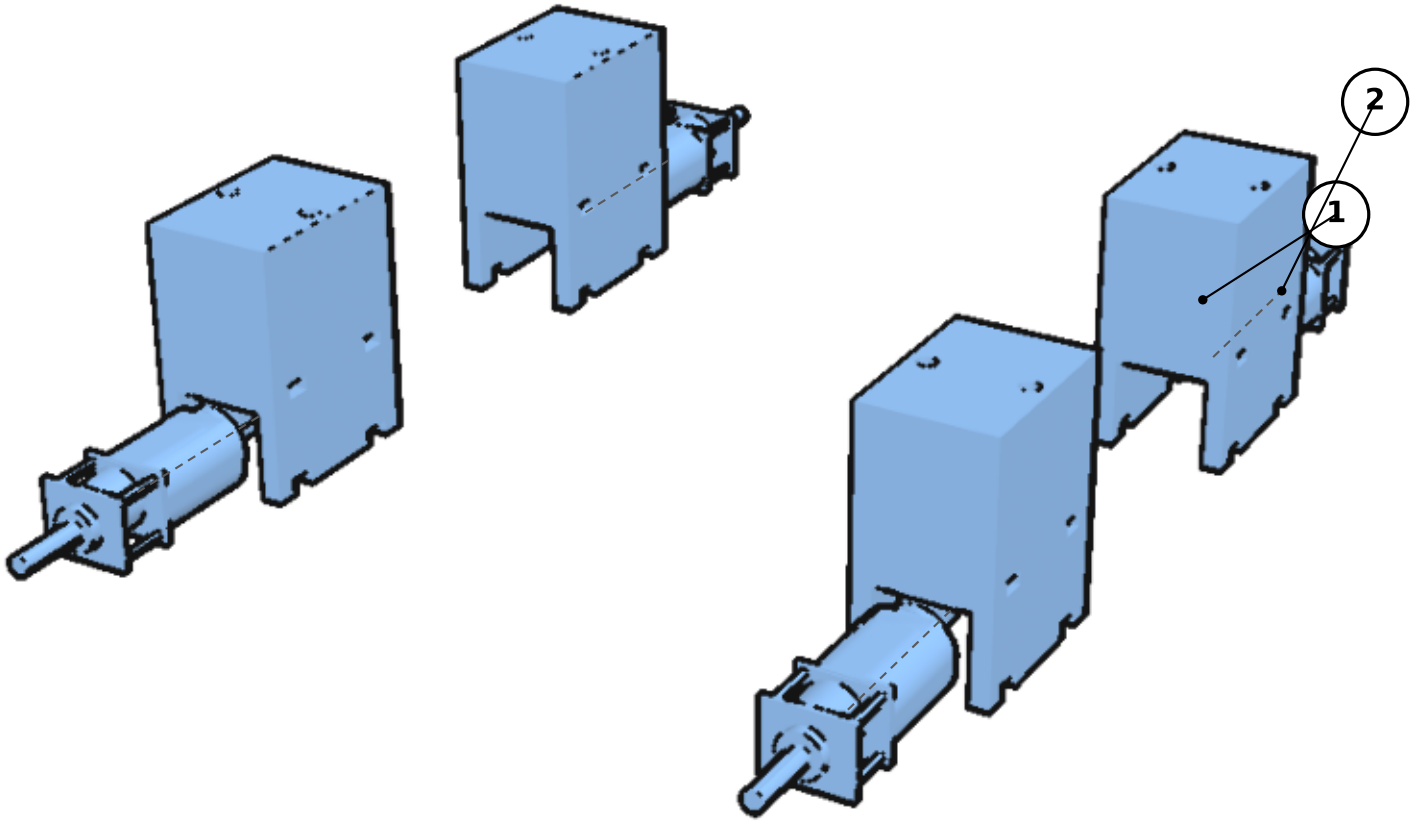
Print first (FABRICATION_GUIDE.md):
laser-cut decks, printed parts, carrier board soldered, headers soldered on every module.

Measure each board you received with calipers and check it fits its engraved outline.

Battery: keep it unplugged until step 6 is wired and both regulators read 5.0 V.

Wiring: AEGIS_Wiring_Guide.pdf (pins, colours, gauges).
First power-up and tests: edge/README.md.

1 Motors into their mounts



PARTS

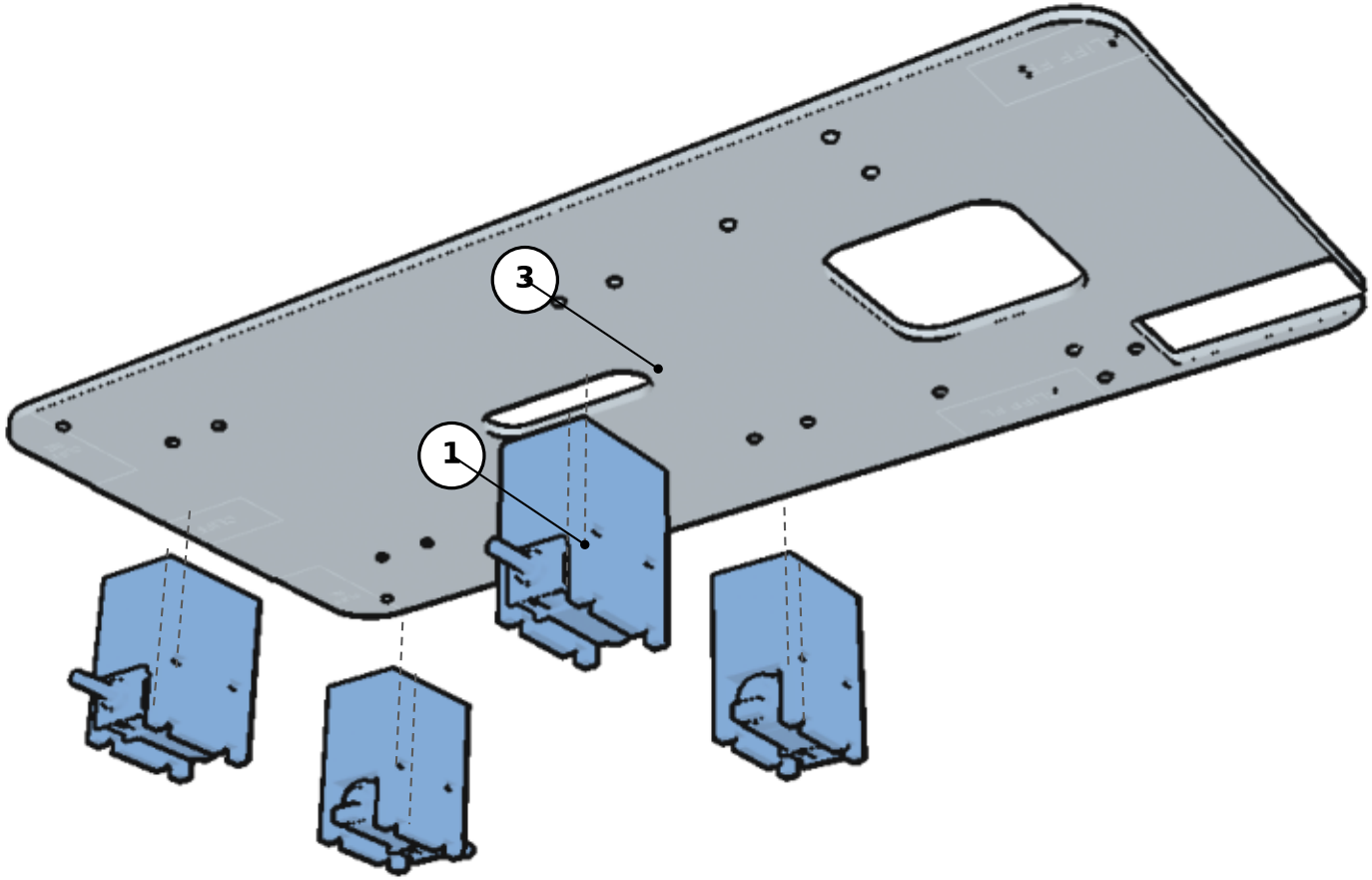
1	Motor mount (printed, black PLA+)	x 4
2	N20 gearmotor with encoder (#5)	x 4
	Cable tie 100 mm (#36)	x 4

Slide each motor into a mount, gearbox end first, encoder cable out of the back.

Loop a cable tie through the two channels in the mount, under the motor, and pull it snug.

Two mounts face left and two face right: the shaft must point outward.

2 Mounts under the lower deck



PARTS

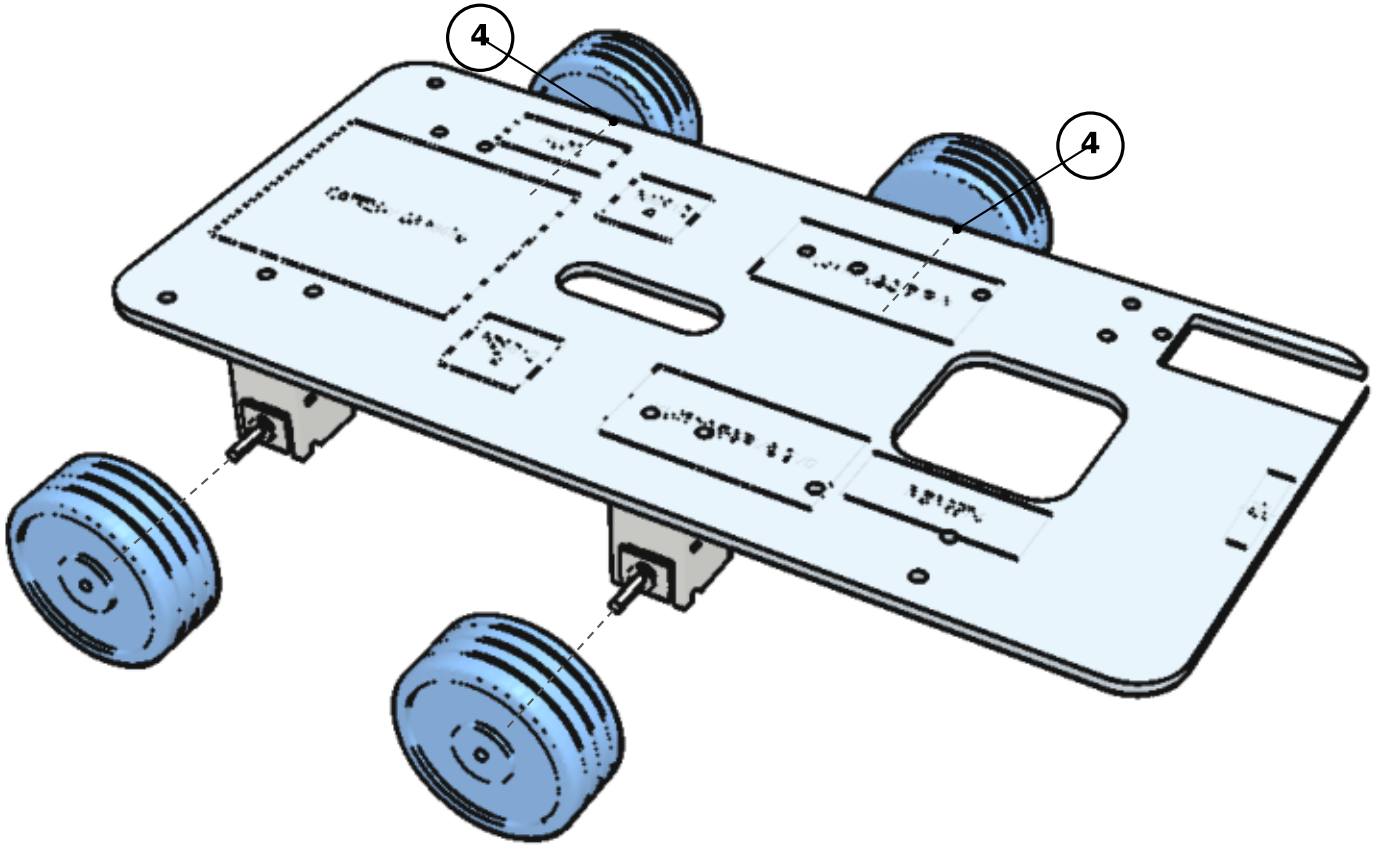
3	Lower deck (laser-cut acrylic, engraved face up)	**x 1**
1	Motor mount with motor	**x 4**
	M3 x 10 screw from the top (#15)	**x 8**

Peel the deck film. Engraved marks face up.

Each mount sits under its two holes; screw M3 x 10 down from the top into the 2.5 mm holes.

Do not over-tighten: the screws cut their own thread in the PLA.

3 Wheels



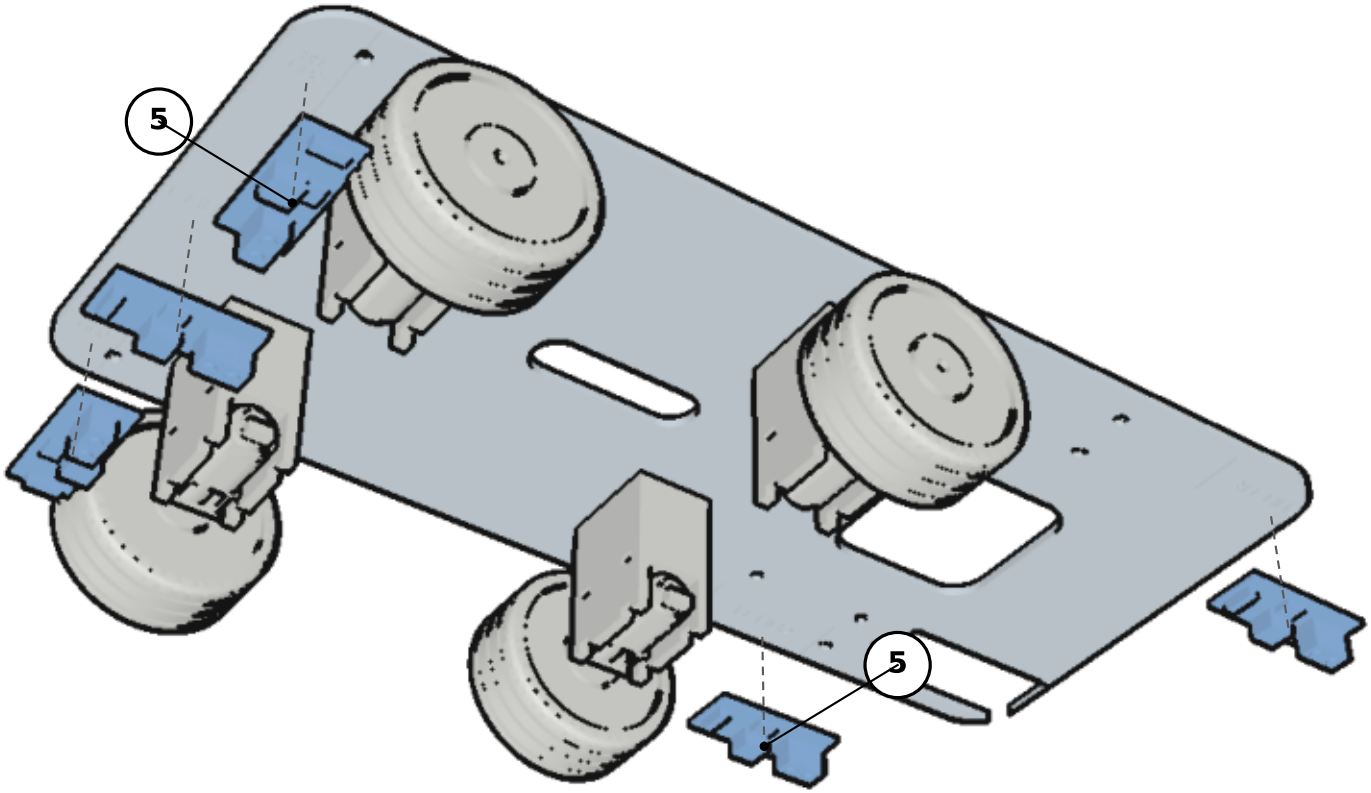
PARTS

4	44 mm N20 wheel (#13)	**x 4**

Press each wheel onto its motor's D-shaft, flat side matching, until it stops.

Spin each wheel by hand: nothing should rub on the mount or the deck.

4 Table-edge sensors under the deck



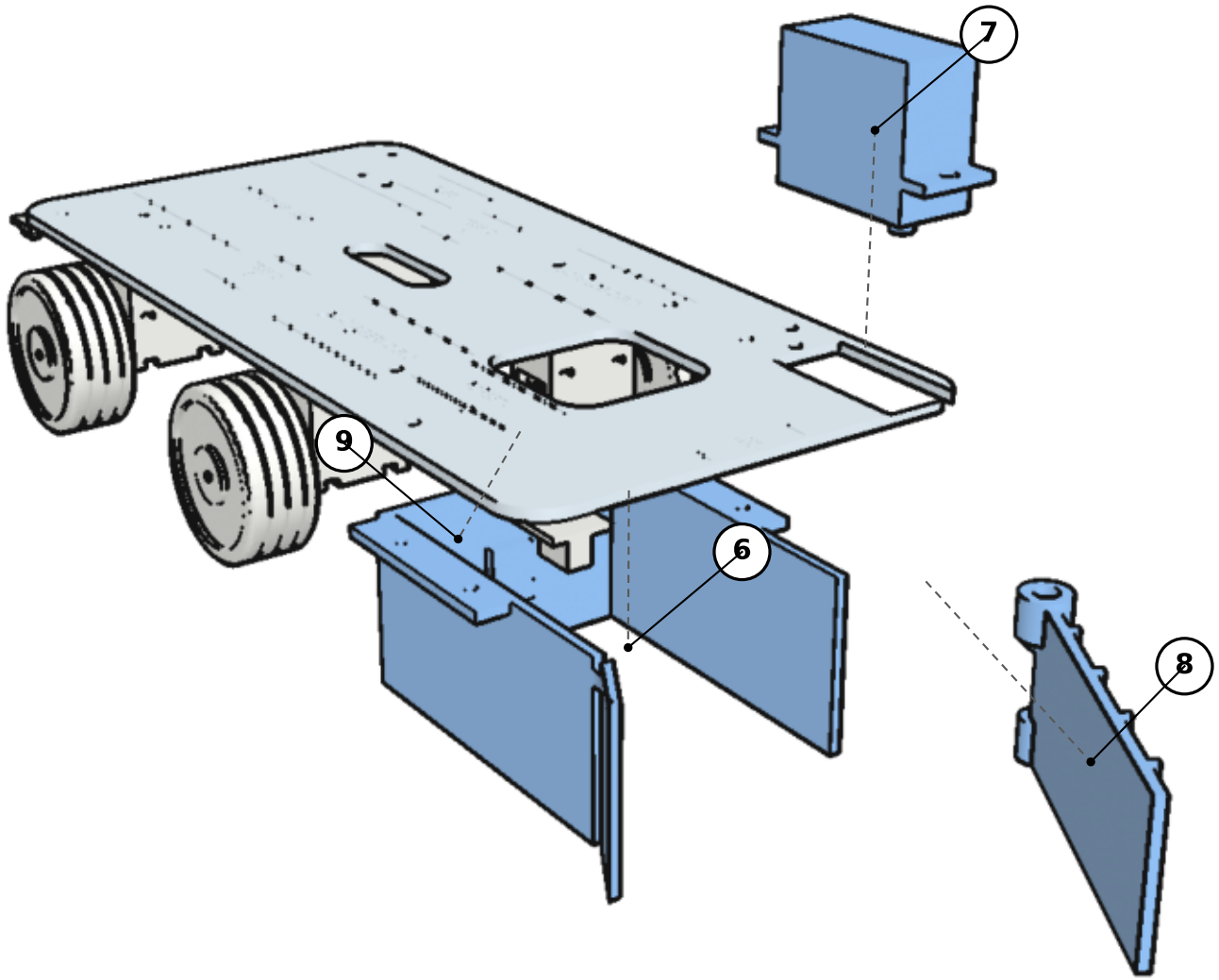
PARTS

5	TCRT5000 edge sensor (#18)	**x 5**
	Foam tape (#38), paper template	

Tape the printed underside template to the deck and stick each sensor on its outline, lens pointing down: front-left, front-right, rear, and both rear corners.

Lens 3-8 mm above the table. Run the wires up through the harness slot.

5 Intake bay: shroud, servo, door, bay sensor

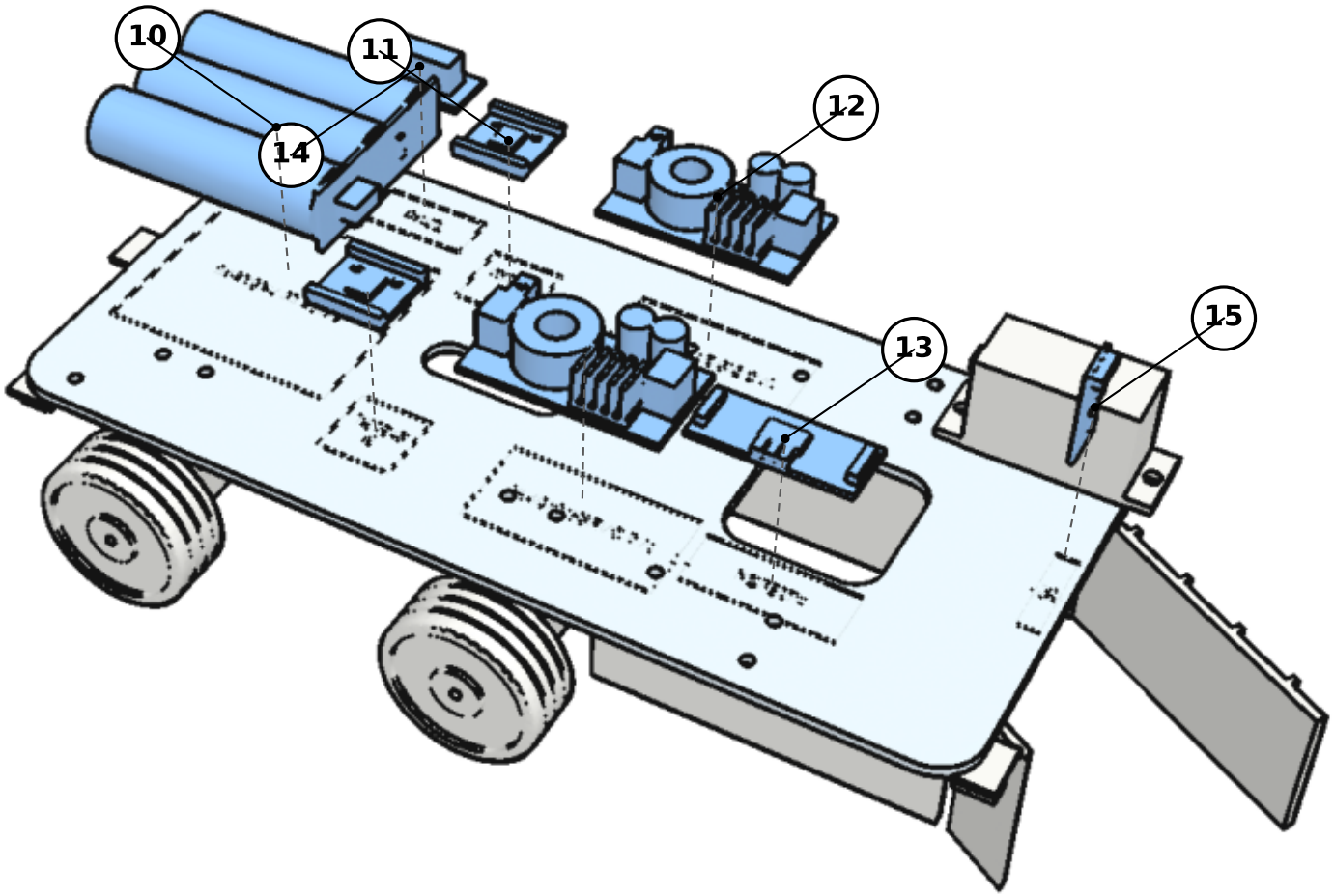


PARTS

6	Intake shroud (printed)	x 1
7	MG946R servo (#9)	x 1
8	Intake door (printed)	x 1
9	IR obstacle sensor (#19)	x 1
	M3 x 10 screw + nut	x 6

Bolt the shroud flanges under the deck (4 x M3). Drop the servo through its pocket, spline down, and screw its ears to the deck (2 x M3). Warm the spline and press the door hub onto it with the servo at the CLOSED position. Tape the bay sensor to the boss on the back wall.

6 Lower-deck electronics



PARTS

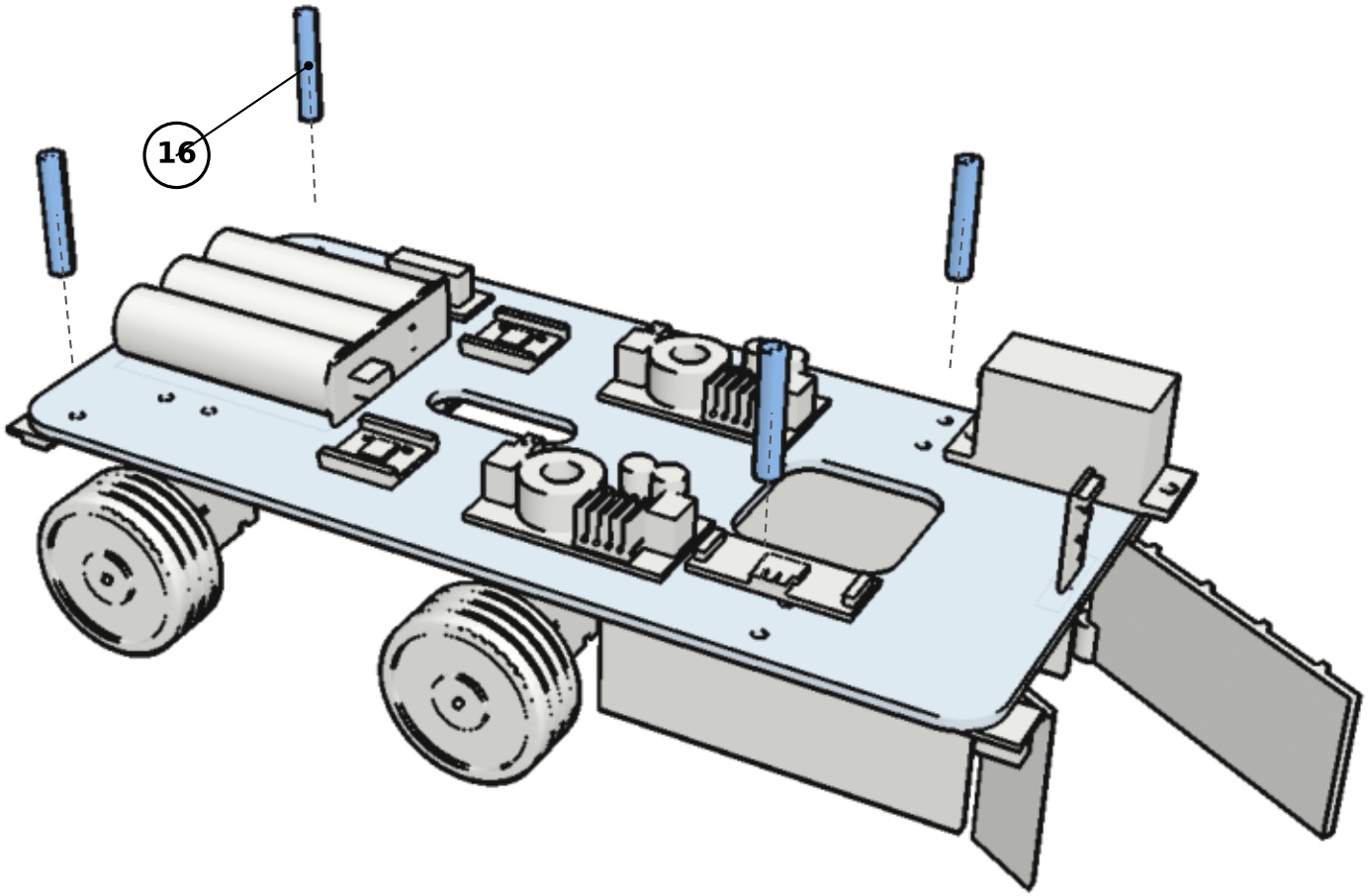
10	3S 18650 pack (#10), cable-tied down	x 1
11	TB6612FNG driver (#6)	x 2
12	XL4015 buck, set to 5.0 V first (#12)	x 2
13	PCF8574 I/O expander (#17)	x 1
14	Fuse board: holder + 6.3 A fuse (#22, #23)	x 1
15	VL53L0X front + L-bracket	x 1

Stick each board on its engraved outline with foam tape; strap the battery with two cable ties.

Set both XL4015 outputs to 5.0 V with a multimeter BEFORE connecting anything to them.

Wire this deck now, following AEGIS_Wiring_Guide.pdf; leave the battery unplugged.

7 Pillars

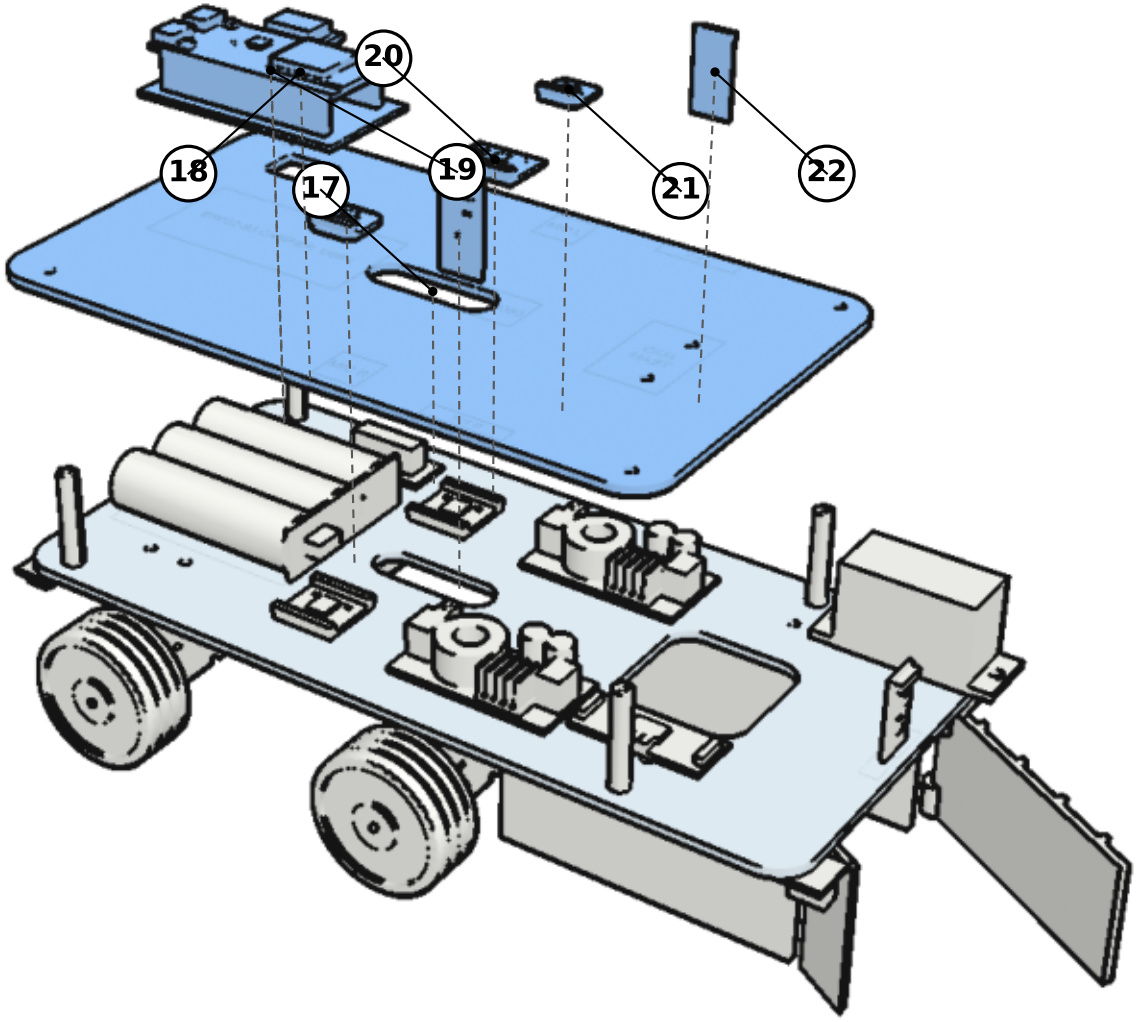


PARTS

16	M3 pillar stack: 20 mm F-F + 10 mm M-F (#15)	**x 4**

Screw a 10 mm M-F pillar up through each corner hole of the lower deck, then a 20 mm F-F pillar on top: 30 mm between the decks.

8 Upper deck



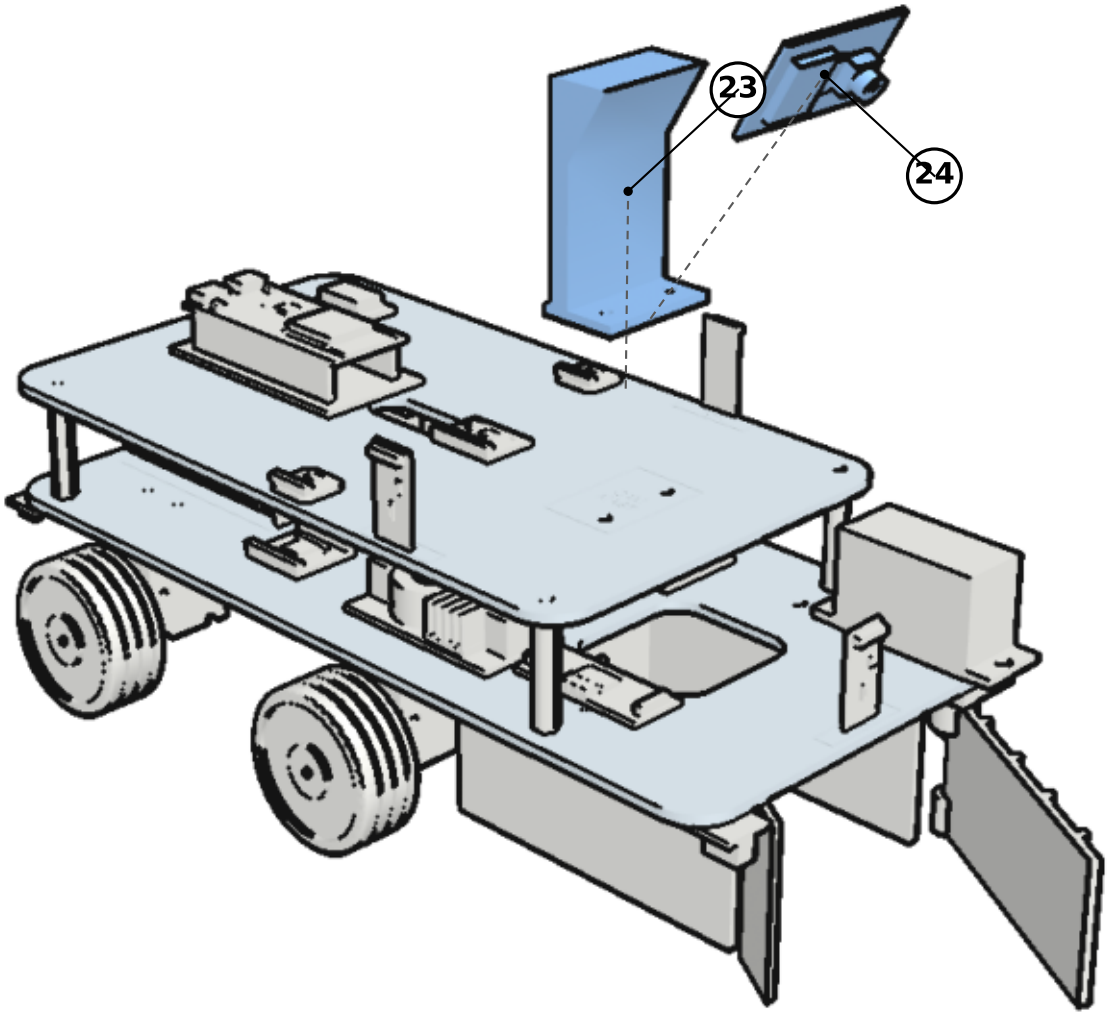
PARTS

17	Upper deck (acrylic)	x 1
18	Power switch, snaps into its cut-out (#21)	x 1
19	ESP32-S3 on its soldered carrier board (#1, #31, #32)	x 1
20	MPU-6050 IMU (#8)	x 1
21	INMP441 microphone, 100 mm apart (#4)	x 2
22	VL53L0X side sensor + L-bracket (#7)	x 2
	M3 x 6 screw	x 4

Feed the harness through the slot, set the deck on the pillars, screw down (4 x M3 x 6).

Press the switch in from the top. Tape the carrier, IMU, mics and side sensors on their outlines; the IMU must sit flat and square to the deck edge. Then wire the upper deck.

9 Camera mast



PARTS

23	Camera mast (printed)	**x 1**
24	ESP32-CAM with OV2640 (#2)	**x 1**
	M3 x 10 screw + nut, foam tape	**x 2**

Bolt the mast to the upper deck (2 x M3 up from below, nuts on the base).

Stick the ESP32-CAM on the sloped pad with two layers of foam tape, lens forward and up, SD slot clear. It looks 30 degrees down at the table.